4.4 Congruence and Transformations

Name _____ Period ____ Date _____

Geometry 4.4 · Student sheet

I can…

- describe a congruence transformation that maps one figure onto another.
- decide whether two figures are congruent.
- use the theorems about reflections in parallel and intersecting lines.

Vocabulary · fill in the missing word

Congruent figures — figures with the same size and _____

Congruence transformation — a transformation that _____ length and angle measure (a translation, reflection, rotation, or a composition of them)

Part 1 · Discovery: get from here to there

Move △ABC with A(−4, 1), B(−2, 1), C(−4, 3) onto △PQR with P(4, −3), Q(2, −3), R(4, −1).

Move	Description (vector, line, or angle and center)
1	
2	
3 (if needed)	

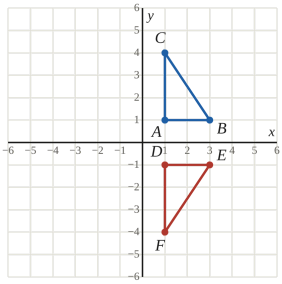
Did the other pair use the same moves?

Part 2 · Notes and examples (fill in each note, then work the example)

Congruence: two figures are congruent if and only if there is a _____ of rigid motions that maps one onto the other.

Example 1

Describe a congruence transformation that maps △ABC to △DEF: A(1, 1), B(3, 1), C(1, 4) and D(1, −1), E(3, −1), F(1, −4).



Example 2

Describe a congruence transformation that maps △ABC with A(−4, 1), B(−2, 1), C(−4, 3) onto △PQR with P(4, −3), Q(2, −3), R(4, −1).

Example 3

Is a 2-by-2 square congruent to a 2-by-3 rectangle? Explain.

Reflections in Parallel Lines Theorem: reflecting in two parallel lines is the same as a _____, twice the distance between the lines.

Example 4

Reflect (0, 2) in the line x = 1, then reflect the image in x = 4. What single transformation does the same thing?

Reflections in Intersecting Lines Theorem: reflecting in two intersecting lines is the same as a _____ about their intersection by twice the angle between them.

Example 5

Two lines intersect at point P and form a 40° angle. A figure is reflected in one line and then the other. What single transformation does the same thing?

Part 3 · Partner practice

Partner 1

Describe a congruence transformation that maps A(2, 3), B(5, 3), C(2, 1) to A′(−3, 2), B′(−3, 5), C′(−1, 2).

Partner 2

A figure is reflected in x = −1 and then in x = 2. Describe the single transformation.